Effective-temperature multipoles for relativistic transport. III. Entropy geometry and tangency structure of the two-field manifold

Jiwon Park,[1,*] Myung-Ki Cheoun,[1,†] and Jubin Park[1,‡]

[1] *Department of Physics and OMEG Institute, Soongsil University, Dongjak-gu, Seoul 06978, Republic of Korea*

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In the companion papers [Papers I and II], we constructed the effective-temperature multipole manifold with a direction-dependent chemical potential and established non-redundancy of the five-parameter ($L_\Theta = 2$, $L_\eta = 1$) state vector for all three quantum statistics $\xi \in \{0, -1, +1\}$. The present paper develops the entropy geometry and tangency structure of this two-field manifold. We prove that the 2×2 Gram matrix $\mathbf{G}_\xi$ of the tangent-space inner product is identically the negative of the entropy second variation restricted to the manifold tangent space, for all three statistics simultaneously (Theorem 1, Gram–Hessian identity). This immediately yields strict entropy concavity on the extended manifold as a corollary (Corollary 2). We show that the tangency diagnostic of Paper I survives on the two-field manifold: the diagnostic vanishes if and only if the collision source lies in the two-dimensional tangent space (Theorem 3). The Pythagorean decomposition (Proposition 5) partitions the one-field diagnostic exactly into an η-tangent component and a genuine spectral-distortion residual; at moderate Fermi–Dirac degeneracy, the η-tangent component accounts for 98.5% of the one-field signal. We establish that the ray-wise thermodynamic Hessian is positive definite for both number-density and energy-density weightings (Proposition 7), and show that the linearity of the angular parameterisation lifts this to positive definiteness of the full 5×5 angular Hessian (Proposition 8), completing the thermodynamic convexity groundwork for the Godunov formulation. We characterise the principal angle between the one-field and two-field tangent spaces (Proposition 6). All results are proved for Maxwell–Boltzmann, Fermi–Dirac, and Bose–Einstein statistics, with the statistics dependence entering solely through the inner-product weight $w_* = f(1+\xi f)\, x^p$. The Gram matrix simultaneously serves as identifiability metric (Paper II), entropy Hessian (Theorem 1), and tangency projection kernel (Theorem 3)—a triple role rooted in the exponential-family structure of the per-ray distribution.

I. INTRODUCTION

Paper I [1] constructed the one-field effective-temperature manifold and equipped it with a tangency diagnostic and an entropy-projection inequality. Paper II [2] extended the manifold to include a direction-dependent chemical potential $\eta(\hat{e})$ and established, for all three quantum statistics, that the five-parameter moment-map Jacobian is full-rank, the scalar chemical potential is insufficient, and the number-to-energy flux fraction ratio $f_\mathcal{N}/f_\mathcal{E}$ is the cleanest observable signature of the η-dipole.

Paper II establishes identifiability but leaves open the entropy structure. The present paper fills this gap. We address three questions for all $\xi \in \{0, -1, +1\}$:

(i) *Does the entropy remain strictly concave on the extended manifold?* We show yes: the entropy Hessian restricted to the tangent space is identical to the Gram matrix that certifies identifiability (Theorem 1). This Gram–Hessian identity is the structural core of the paper.

(ii) *Does the tangency diagnostic survive on the two-field manifold?* We show yes: the diagnostic detects genuine off-manifold content with a precisely characterized reduction from the one-field case (Theorem 3, Propositions 4 and 5).

(iii) *Does the thermodynamic potential remain convex?* We establish the ray-wise positivity of the two-field Hessian for both number-density and energy-density weightings (Proposition 7).

Scope. All formal results (Theorems 1 and 3, Corollary 2, Propositions 4–8) are proved analytically for all three quantum statistics. The diagnostic reduction factors (Table II) are numerical over tested configurations. We restrict to the ($L_\Theta = 2$, $L_\eta = 1$) truncation and tested collision operators (multipole relaxation and BGK). We do not implement a transport solver, present astrophysical benchmarks, or prove the full symmetric hyperbolicity of the PDE system.

II. ENTROPY STRUCTURE ON THE EXTENDED MANIFOLD

A. Motivation and setup

In moment-based kinetic theory, the choice of closure determines the thermodynamic consistency of the reduced system. A closure is thermodynamically consistent if the reduced entropy—the entropy evaluated on the closure manifold—is a concave function of the closure parameters; concavity ensures that the reduced system inherits the dissipative character of the underlying Boltz-

* cosmosapjw@soongsil.ac.kr
† cheoun@ssu.ac.kr
‡ holonov@ssu.ac.kr

mann equation and does not admit entropy-increasing perturbations [3, 4]. Paper I [1] established this concavity for the one-field Teff manifold (parameterised by Θ alone) by identifying the per-ray entropy Hessian with the Laguerre inner product. Paper II [2] extended the manifold to include a direction-dependent chemical potential $\eta(\hat{e})$, proving that the five-parameter state vector is identifiable through a positive-definite 2×2 Gram matrix $\mathbf{G}_\xi$. What Paper II did not address is whether the extended manifold inherits the entropy concavity of the one-field case. The present section shows that it does, and that the Gram matrix certifying identifiability is *identically* the entropy Hessian restricted to the tangent space. This is not a coincidence: it reflects the exponential-family structure of the per-ray distribution.

We work with the two-field distribution introduced in Paper II [2]:

$$f(x;\Theta,\eta) = \Phi_\xi\Big(\frac{x}{\Theta}-\eta\Big) = \Big[\exp\Big(\frac{x}{\Theta}-\eta\Big)-\xi\Big]^{-1}, \quad (1)$$

where $x = E/(k_{\mathrm{B}}T_0) > 0$ is the direction-independent dimensionless energy (Paper II, Theorem 1), $\Theta(\hat{e}) > 0$ is the direction-dependent effective temperature, $\eta(\hat{e})$ is the direction-dependent degeneracy parameter, and $\xi \in \{0,-1,+1\}$ selects the quantum statistics. The inner product used throughout is

$$\langle g,h\rangle_* := \int_0^\infty g(x)\,h(x)\,f(1+\xi f)\,x^3\,\mathrm{d}x, \quad (2)$$

with weight $w_* = f(1+\xi f)\,x^3 > 0$ for all $x>0$, $\Theta>0$, and admissible η. This is the Fisher-metric weight: it arises naturally from both the information geometry and the entropy geometry, as we now show.

B. The entropy generating function

For each statistics $\xi \in \{0,-1,+1\}$, the Boltzmann–Gibbs entropy generating function is

$$\sigma_\xi(f) = -f\ln f - \xi^{-1}(1+\xi f)\ln(1+\xi f), \quad (3)$$

where the $\xi = 0$ (MB) case is defined by continuity: $\lim_{\xi\to 0}\xi^{-1}(1+\xi f)\ln(1+\xi f) = f$, giving $\sigma_\xi[0](f) = -f\ln f - f$. whose first and second derivatives are

$$\sigma'_\xi(f) = -\ln\frac{f}{(1+\xi f)^{1/\xi}}, \qquad \sigma''_\xi(f) = -\frac{1}{f(1+\xi f)}. \quad (4)$$

The strict negativity of σ''_ξ for all physical f ($f>0$; for FD, $f<1$; for BE, $f>0$) is the local expression of the H-theorem: any perturbation of the distribution at fixed energy increases the entropy generating function. For the three statistics explicitly:

$$\begin{aligned}
\text{MB }(\xi{=}0):&\quad \sigma_0 = -f\ln f + f,\quad \sigma''_0 = -1/f, && (5)\\
\text{FD }(\xi{=}{-1}):&\quad \sigma_{-1} = -f\ln f - (1-f)\ln(1-f), &&\\
&\quad \sigma''_{-1} = -1/[f(1-f)], && (6)\\
\text{BE }(\xi{=}{+1}):&\quad \sigma_{+1} = -f\ln f + (1+f)\ln(1+f), &&\\
&\quad \sigma''_{+1} = -1/[f(1+f)]. && (7)
\end{aligned}$$

For FD, the weight $f(1-f)$ is the variance of a Bernoulli random variable with parameter f: it peaks at $f = 1/2$ and vanishes at $f=0$ and $f=1$ (Pauli blocking). For BE, the weight $f(1+f)$ grows without bound as $f\to\infty$ (Bose enhancement). For MB, the weight is simply f: there is no quantum correction.

The total entropy functional for a single angular direction is

$$S_\xi[f] = \int_0^\infty \sigma_\xi\big(f(x)\big)\,x^3\,\mathrm{d}x, \quad (8)$$

where the factor x^3 is the energy-density phase-space measure. The entropy S_ξ is strictly concave because σ_ξ is strictly concave and $x^3>0$.

C. Second variation on the tangent space

We now compute the second variation of S_ξ along the tangent space of the (Θ,η) manifold. The derivation has three steps: (i) identify the tangent vectors, (ii) compute $(\delta f)^2$ in terms of the tangent-space coordinates, and (iii) evaluate the integral, exploiting a key cancellation between σ''_ξ and $(\delta f)^2$.

Step 1: tangent vectors. The partial derivatives of (1) with respect to the two parameters are

$$\partial_\Theta f = f(1+\xi f)\,\frac{x}{\Theta^2}, \qquad \partial_\eta f = f(1+\xi f). \quad (9)$$

Both tangent vectors share the common factor $f(1+\xi f)$, which is the Fisher-metric weight. We define the normalised tangent-direction functions $v_1(x) := x/\Theta^2$ and $v_2(x) := 1$, so that $\partial_\Theta f = f(1+\xi f)\,v_1$ and $\partial_\eta f = f(1+\xi f)\,v_2$. The function v_1 is strictly increasing in x (it measures the spectral slope response to a temperature perturbation), while v_2 is constant (it measures the uniform-shift response to a chemical potential perturbation). Their non-proportionality is the geometric origin of parameter identifiability (Paper II, Proposition 2).

A tangent perturbation $\delta\boldsymbol{\alpha} = (\delta\Theta,\delta\eta)$ induces a distribution change

$$\delta f = \partial_\Theta f\;\delta\Theta + \partial_\eta f\;\delta\eta = f(1+\xi f)\big(v_1\,\delta\Theta + v_2\,\delta\eta\big). \quad (10)$$

Step 2: the key cancellation. The second variation of S_ξ is

$$\delta^2 S_\xi = \int_0^\infty \sigma''_\xi(f)\,(\delta f)^2\,x^3\,\mathrm{d}x. \quad (11)$$

Substituting $\sigma''_\xi(f) = -1/[f(1+\xi f)]$ from (4) and δf from (10):

$$\begin{aligned}
\sigma''_\xi(f)\,(\delta f)^2 &= \frac{-1}{f(1+\xi f)}\cdot\big[f(1+\xi f)\big]^2\big(v_1\delta\Theta + v_2\delta\eta\big)^2\\
&= -f(1+\xi f)\big(v_1\delta\Theta + v_2\delta\eta\big)^2. \quad (12)
\end{aligned}$$

The $[f(1+\xi f)]^2$ from $(\delta f)^2$ cancels against the $1/[f(1+\xi f)]$ from σ''_ξ, leaving a single power of the Fisher weight. This cancellation is the hallmark of exponential families: the curvature of the log-likelihood (Fisher information) equals the curvature of the entropy (information-geometric duality). For MB, the cancellation is $(-1/f)\cdot f^2 = -f$; for FD, $(-1/[f(1-f)])\cdot[f(1-f)]^2 = -f(1-f)$; for BE, $(-1/[f(1+f)])\cdot[f(1+f)]^2 = -f(1+f)$. In all three cases the surviving weight is $f(1+\xi f)$, which is precisely the inner-product weight w_* in (2).

Step 3: the Gram matrix. Inserting (12) into (11) and expanding the square:

$$\begin{aligned}\delta^2 S_\xi &= -\int_0^\infty f(1+\xi f)\big(v_1\delta\Theta + v_2\delta\eta\big)^2 x^3\,\mathrm{d}x\\ &= -\big[\langle v_1,v_1\rangle_*\,(\delta\Theta)^2 + 2\langle v_1,v_2\rangle_*\,\delta\Theta\,\delta\eta\\ &\quad + \langle v_2,v_2\rangle_*\,(\delta\eta)^2\big]\\ &= -\delta\boldsymbol{\alpha}^T\,\mathbf{G}_\xi\,\delta\boldsymbol{\alpha},\end{aligned}\tag{13}$$

where the Gram matrix components are

$$[\mathbf{G}_\xi]_{11} = \langle v_1,v_1\rangle_* = \int_0^\infty f(1+\xi f)\,\frac{x^2}{\Theta^4}\,x^3\,\mathrm{d}x,\tag{14}$$

$$[\mathbf{G}_\xi]_{12} = \langle v_1,v_2\rangle_* = \int_0^\infty f(1+\xi f)\,\frac{x}{\Theta^2}\,x^3\,\mathrm{d}x,\tag{15}$$

$$[\mathbf{G}_\xi]_{22} = \langle v_2,v_2\rangle_* = \int_0^\infty f(1+\xi f)\,x^3\,\mathrm{d}x.\tag{16}$$

These are precisely the components of the 2×2 Gram matrix defined in Paper II (Proposition 2 therein), computed in the inner product (2). The derivation is complete: no approximation has been made at any step.

D. The Gram–Hessian identity

The computation of §II C establishes:

Theorem 1 (Gram–Hessian identity, all ξ)**.** *For the distribution* (1) *at any* $\Theta > 0$*, admissible* η*, and statistics* $\xi \in \{0,-1,+1\}$*, the second variation of the entropy functional* (8) *restricted to the tangent space of the* (Θ,η) *manifold satisfies*

$$\delta^2 S_\xi[\delta\boldsymbol{\alpha}] = -\delta\boldsymbol{\alpha}^T\,\mathbf{G}_\xi(\Theta,\eta)\,\delta\boldsymbol{\alpha},\tag{17}$$

where $\mathbf{G}_\xi$ *is the* 2×2 *Gram matrix* (14)–(16)*. The identity holds exactly, without approximation, for arbitrary tangent perturbation amplitude* $\delta\boldsymbol{\alpha} \neq 0$.

Proof. The three-step derivation of §II C—tangent vectors (9), key cancellation (12), and Gram-matrix expansion (13)—constitutes the proof. The sole ingredients are the entropy second derivative (4), the chain rule, and the definition of the inner product (2). □

Corollary 2 (Strict concavity on the extended manifold, all ξ)**.** *Since* $\mathbf{G}_\xi$ *is positive definite for all* $\Theta > 0$ *and admissible* η *(Paper II [2], Proposition 2; tangent-space independence), the entropy* S_ξ *is strictly concave along all tangent directions:* $\delta^2 S_\xi < 0$ *for all* $\delta\boldsymbol{\alpha} \neq \mathbf{0}$*. No tangent perturbation can increase the entropy.*

E. Significance and scope of the identity

Three aspects of Theorem 1 merit discussion.

First: the dual role. The Gram matrix $\mathbf{G}_\xi$ now serves two distinct functions. In Paper II, it certifies *identifiability*: positive definiteness of $\mathbf{G}_\xi$ means that the parameters Θ and η can be separately recovered from moment data. Here, the same matrix certifies *entropy concavity*: positive definiteness means that no tangent perturbation increases the entropy. These are structurally the same condition. A manifold on which parameters are unidentifiable ($\det\mathbf{G}_\xi = 0$) would also have a flat entropy direction—the two pathologies coincide. This is a nontrivial structural guarantee: identifiability (a statistical-inference property) and thermodynamic consistency (a physical property) are locked together by the exponential-family structure.

Second: statistics universality. The identity (17) holds for all ξ with the same algebraic form. The statistics dependence enters only through the weight $w_* = f(1+\xi f)\,x^3$: for MB, $w_* = f\,x^3$ (Boltzmann weight); for FD, $w_* = f(1-f)\,x^3$ (Pauli-suppressed Fisher weight); for BE, $w_* = f(1+f)\,x^3$ (Bose-enhanced weight). The cancellation (12) works for all three because it relies only on the algebraic identity $\sigma_\xi''(f)\cdot[f(1+\xi f)]^2 = -f(1+\xi f)$, which holds by the definition of σ_ξ.

Third: exponential-family origin. The distribution (1) is a two-parameter exponential family with natural parameters $\boldsymbol{\theta} = (-1/\Theta, \eta)$ and sufficient statistics $\boldsymbol{t} = (x, 1)$. For any exponential family, the Fisher information matrix, the entropy Hessian, and the natural-parameter Gram matrix coincide (see, e.g., Amari [5], Chapter 3). Theorem 1 is a formal instantiation of this general principle for the specific inner product (2). The per-ray tangency diagnostic of Paper I—which projects onto the orthogonal complement of the tangent space in the same inner product—is therefore simultaneously an entropy-curvature diagnostic: it measures how much of the collision source lies outside the directions of maximum entropy decrease.

Generalisation to m fields. The Gram–Hessian identity (Theorem 1) and the tangency criterion (Theorem 3) hold for any finite-dimensional exponential family $f_{\boldsymbol{\vartheta}} = [\exp(\boldsymbol{\vartheta}\cdot\boldsymbol{t}(x)) - \xi]^{-1}$ with m linearly independent sufficient statistics $\boldsymbol{t}$: the $m\times m$ Gram matrix in the inner product $\langle g,h\rangle_* = \int g\,h\,f(1+\xi f)\,w\,\mathrm{d}x$ is simultaneously the entropy Hessian, the Fisher information, and the tangency projection kernel. The two-field results of this paper are the $m = 2$, $\boldsymbol{t} = (x,1)$ specialisation. Extensions to $m > 2$ (e.g. adding energy-shape degrees of freedom via higher sufficient statistics $t_3 = x^2$) inherit the same structural framework.

F. Worked example

To make Theorem 1 concrete, we evaluate the Gram matrix and verify the entropy concavity at $\Theta = 1.2$, $\eta_0 =$

3 for all three statistics.

The Gram matrix at this configuration (Paper II, Table I) has eigenvalues: MB: $\lambda_{\min} = 0.27$, $\lambda_{\max} = 21$, $\kappa = 78$; FD: $\lambda_{\min} = 0.99$, $\lambda_{\max} = 350$, $\kappa = 355$; BE is inadmissible at $\eta_0 = 3$ (requires $\eta_0 \leq 0$); at $\eta_0 = -1$: $\lambda_{\min} = 0.96$, $\lambda_{\max} = 68$, $\kappa = 71$.

Consider the tangent perturbation $\delta\boldsymbol{\alpha} = (0.01, 0.05)$—a 1% temperature shift and 5% chemical-potential shift. By Theorem 1, the entropy change is

$$\delta^2 S_\xi = -(0.01)^2\,[\mathbf{G}_\xi]_{11} - 2(0.01)(0.05)\,[\mathbf{G}_\xi]_{12} - (0.05)^2\,[\mathbf{G}_\xi]_{22}\,.$$

For FD at $(\Theta, \eta_0) = (1.2, 3)$, the Gram components are $[\mathbf{G}_\xi]_{11} \approx 350$, $[\mathbf{G}_\xi]_{12} \approx 18.6$, $[\mathbf{G}_\xi]_{22} \approx 1.36$, giving $\delta^2 S_\xi \approx -(0.035 + 0.019 + 0.0034) = -0.057 < 0$. The dominant contribution comes from $[\mathbf{G}_\xi]_{11}$ (the Θ-Θ component), reflecting the steep entropy curvature along the temperature direction. For MB at the same perturbation: $\delta^2 S_\xi \approx -(0.0021 + 0.0011 + 0.00019) = -0.0034$. The entropy change is 17 times smaller because the MB weight $f\,x^3$ is much lighter than the FD weight $f(1-f)\,x^3$ at moderate degeneracy.

This example illustrates the key physical content of Corollary 2: the entropy decreases for any nonzero tangent perturbation, but the rate of decrease depends strongly on the statistics and the degeneracy level. At high FD degeneracy, the entropy is steeply curved along the Θ-direction (large $[\mathbf{G}_\xi]_{11}$) but nearly flat along the η-direction (small $[\mathbf{G}_\xi]_{22}$): Pauli blocking suppresses the sensitivity to chemical-potential perturbations.

G. Conditioning as entropy curvature

The condition number $\kappa(\mathbf{G}_\xi)$ of the Gram matrix—characterised quantitatively in Paper II (Proposition 9)—acquires a second physical interpretation through Theorem 1: it is the ratio of entropy curvatures along the two tangent directions (Fig. 3, Table I).

For MB ($\xi = 0$): κ is exactly η_0-independent ($\kappa_{\mathrm{MB}} \approx 78$ at $\Theta = 1.2$). The entropy surface has identical curvature ratios at all degeneracies because the common factor e^{η_0} cancels in every ratio of Gram integrals (Paper II, Proposition 16). Geometrically, the MB entropy surface is a fixed paraboloid whose axes scale uniformly with η_0.

For FD ($\xi = -1$): κ increases monotonically with η_0, from 92 at $\eta_0 = 0$ to > 3000 at $\eta_0 = 10$. The physical origin is Pauli blocking: at high degeneracy, the Fisher weight $f(1-f)$ concentrates near the Fermi surface $x \approx \Theta\eta_0$, where both tangent vectors $v_1 = x/\Theta^2$ and $v_2 = 1$ are approximately constant over the narrow Fermi window. The resulting near-collinearity drives the soft eigenvalue toward zero. The soft eigenvector aligns with the η-direction (99.4% alignment at $\eta_0 = 10$; Paper II, Table IV), confirming that the nearly flat entropy direction is the chemical-potential direction. Physically: at high degeneracy, shifting η at fixed Θ barely changes the distribution because most states are already occupied; only modes near the Fermi surface respond, and their contribution to the w_*-weighted integral is progressively smaller.

For BE ($\xi = +1$): $\kappa < \kappa_{\mathrm{MB}}$ throughout the admissible domain $\eta_0 \leq 0$; $\kappa \approx 63$ at $\eta_0 = -0.3$. Bose enhancement $f(1+f)$ broadens the effective support of w_* and increases the functional separation between v_1 and v_2, improving the entropy-curvature separation. In the deep bulk ($\eta_0 \to -\infty$), $f \ll 1$ and $(1+f) \to 1$, so $\kappa_{\mathrm{BE}} \to \kappa_{\mathrm{MB}}$ from below.

The conditioning does not break strict concavity: both eigenvalues remain strictly positive at all admissible parameters (Corollary 2). High conditioning is a practical caveat for numerical inversion (entropy-based Newton methods converge slowly along the soft direction), not a structural obstruction.

III. TANGENCY DIAGNOSTIC ON THE TWO-FIELD MANIFOLD

A. Motivation

The tangency diagnostic of Paper I detects whether collisions drive the distribution away from the effective-temperature manifold. On the original one-field manifold parameterised by Θ alone, the tangent space at each direction is one-dimensional: $V_1 = \mathrm{span}\{v_1\}$ with $v_1 = x/\Theta^2$ in the $\langle\cdot,\cdot\rangle_*$ inner product. The diagnostic $\mathcal{D}^{(1\mathrm{D})}_{\geq 2}$ measures the norm of the collision-source component that lies outside this one-dimensional subspace. A nonzero residual signals that the collision operator generates energy-shape content—spectral distortions in the variable x—that cannot be absorbed by adjusting Θ.

On the extended (Θ, η) manifold, the tangent space is two-dimensional: $V_2 = \mathrm{span}\{v_1, v_2\}$ with $v_1 = x/\Theta^2$ (temperature direction) and $v_2 = 1$ (chemical-potential direction). The additional basis function v_2 corresponds to the $\partial_\eta f$ direction—perturbations that shift the overall occupation level without changing the spectral slope. Two questions arise: (i) does the extended diagnostic still detect genuine off-manifold content, or does the larger tangent space absorb everything? (ii) by how much does the diagnostic value decrease when the tangent space is enlarged?

The answers—Theorem 3 for (i) and Propositions 4 and 5 for (ii)—are geometrically immediate but physically consequential. The key result is the Pythagorean decomposition (Proposition 5): the one-field diagnostic partitions exactly into an η-tangent component (absorbed by the two-field manifold) and a genuine spectral-distortion residual. At moderate Fermi–Dirac degeneracy, the η-tangent component accounts for 98.5% of the one-field signal—the one-field diagnostic overwhelmingly misclassifies chemical-potential content as off-manifold spectral distortion.

B. Diagnostic definition

Let $C[f](x)$ denote the collision operator evaluated at the distribution $f(x;\Theta,\eta)$ along a given angular direction. The entropy-normalised collision source is

$$G_\xi(x) = \frac{C[f](x)}{f(x)(1{+}\xi f(x))}\,, \tag{18}$$

where the denominator $f(1{+}\xi f)$ is the same Fisher weight that appears in the inner product (2) and the Gram–Hessian identity (Theorem 1). The normalisation converts the collision integral from the distribution space to the natural-parameter space of the exponential family. Physically, $G_\xi(x)$ measures the "force" that collisions exert on the natural parameters $(-1/\Theta,\eta)$, projected onto the energy variable x.

The orthogonal projection of G_ξ onto the tangent subspace V_n in $\langle\cdot,\cdot\rangle_*$ is

$$\Pi_n G_\xi = \sum_{i,j=1}^{n} [\mathbf{G}_\xi^{-1}]_{ij}\,\langle G_\xi, v_j\rangle_*\, v_i\,, \tag{19}$$

where $[\mathbf{G}_\xi]_{ij} = \langle v_i, v_j\rangle_*$ is the Gram matrix and $\{v_i\}$ is a basis of V_n. The diagnostic residual is

$$\begin{aligned} D_{\geq n+1} &= \|G_\xi - \Pi_n G_\xi\|_* \\ &= \sqrt{\langle G_\xi - \Pi_n G_\xi,\; G_\xi - \Pi_n G_\xi\rangle_*}\,. \end{aligned} \tag{20}$$

For the one-field manifold: $V_1 = \mathrm{span}\{v_1\}$ with $v_1 = x/\Theta^2$, giving $\mathcal{D}_{\geq 2}^{(1\mathrm{D})}$. For the two-field manifold: $V_2 = \mathrm{span}\{v_1, v_2\}$ with $v_2 = 1$, giving $\mathcal{D}_{\geq 2}^{(2\mathrm{D})}$.

C. Detection theorem

Theorem 3 (Tangency diagnostic on the two-field manifold, all ξ)**.** *Let $G_\xi \in L^2(w_*)$ be an entropy-normalised collision source. The extended diagnostic satisfies*

$$\mathcal{D}_{\geq 2}^{(2\mathrm{D})} = 0 \quad\Longleftrightarrow\quad G_\xi \in V_2 = \mathrm{span}\Big\{\frac{x}{\Theta^2},\, 1\Big\} \;\; \text{in } \langle\cdot,\cdot\rangle_*. \tag{21}$$

That is, the diagnostic vanishes if and only if the collision source is tangent to the per-ray (Θ,η) family at the current state. As in Paper I, this is a per-ray, instantaneous, collision-side statement; tangency to the globally truncated (L_Θ, L_η) angular manifold additionally requires angular-consistency of the constant modes across directions (Paper I, Remark V B).

Proof. The inner product $\langle\cdot,\cdot\rangle_*$ is positive definite because $w_* = f(1{+}\xi f)\,x^3 > 0$ for all $x > 0$ and admissible (Θ,η,ξ). Hence $L^2(w_*)$ is a Hilbert space, and the orthogonal projection $\Pi_2 : L^2(w_*) \to V_2$ is well-defined. The residual $G_\xi - \Pi_2 G_\xi$ lies in $V_2^\perp$ (the orthogonal complement of V_2 in $L^2(w_*)$). By the Hilbert-space projection theorem:

$$\begin{aligned} &\|G_\xi - \Pi_2 G_\xi\|_* = 0 \\ &\quad\Longleftrightarrow\quad G_\xi - \Pi_2 G_\xi = 0 \\ &\quad\Longleftrightarrow\quad G_\xi = \Pi_2 G_\xi \in V_2. \end{aligned}$$

The second equivalence uses the fact that $\|\cdot\|_*$ is a norm (positive definiteness of w_* ensures $\|h\|_* = 0$ iff $h = 0$ a.e. with respect to $w_*\,\mathrm{d}x$). □

Physical content. The theorem sharpens the diagnostic interpretation relative to Paper I. On the one-field manifold, $\mathcal{D}_{\geq 2}^{(1\mathrm{D})} = 0$ means the collision source is proportional to $\partial_\Theta f$—it can be absorbed by a temperature adjustment alone. On the two-field manifold, $\mathcal{D}_{\geq 2}^{(2\mathrm{D})} = 0$ means the source lies in the two-dimensional span of $\partial_\Theta f$ and $\partial_\eta f$—it can be absorbed by adjusting *both* Θ and η. A nonzero $\mathcal{D}_{\geq 2}^{(2\mathrm{D})}$ signals genuine spectral distortions—energy-shape content beyond what temperature and chemical-potential shifts can represent. For a multipole-relaxation operator that drives $(\Theta,\eta) \to (\Theta_{\mathrm{iso}}, \eta_{\mathrm{iso}})$ via a linear combination of $\partial_\Theta f$ and $\partial_\eta f$, the source G_ξ lies in V_2 by construction, so $\mathcal{D}_{\geq 2}^{(2\mathrm{D})} = 0$ to quadrature precision ($< 10^{-12}$ at all tested configurations). For a BGK isotropisation operator $C = (f_{\mathrm{iso}} - f)/\tau$, the source generically contains energy-mode content in $x^2, x^3, \ldots$ that cannot be captured by V_2; accordingly $\mathcal{D}_{\geq 2}^{(2\mathrm{D})} > 0$ at all tested configurations (Table II).

D. Monotone reduction

The two-field tangent space V_2 contains the one-field tangent space V_1 as a subspace ($V_1 \subset V_2$). Projecting onto a larger subspace can only reduce the residual.

Proposition 4 (Monotone reduction, all ξ)**.** $\mathcal{D}_{\geq 2}^{(2\mathrm{D})} \leq \mathcal{D}_{\geq 2}^{(1\mathrm{D})}$*, with equality if and only if G_ξ has no component along the η-direction orthogonal to V_1.*

Proof. Since $V_1 \subset V_2$, the orthogonal complements satisfy $V_2^\perp \subset V_1^\perp$. Decompose $G_\xi = \Pi_1 G_\xi + (G_\xi - \Pi_1 G_\xi)$. The residual $G_\xi - \Pi_1 G_\xi \in V_1^\perp$ can be further decomposed as the sum of a component in $V_2 \ominus V_1$ (the orthogonal complement of V_1 within V_2) and a component in $V_2^\perp$. Since these two components are orthogonal:

$$\|\Pi_{V_1^\perp} G_\xi\|_*^2 = \|\Pi_{V_2 \ominus V_1} G_\xi\|_*^2 + \|\Pi_{V_2^\perp} G_\xi\|_*^2. \tag{22}$$

The left side is $[\mathcal{D}_{\geq 2}^{(1\mathrm{D})}]^2$ and the last term is $[\mathcal{D}_{\geq 2}^{(2\mathrm{D})}]^2$. Since $\|\Pi_{V_2 \ominus V_1} G_\xi\|_*^2 \geq 0$, we have $[\mathcal{D}_{\geq 2}^{(2\mathrm{D})}]^2 \leq [\mathcal{D}_{\geq 2}^{(1\mathrm{D})}]^2$. Equality holds iff $\Pi_{V_2 \ominus V_1} G_\xi = 0$, i.e. iff G_ξ has no component in the η-direction orthogonal to V_1. □

Physical interpretation. The inequality is a geometric necessity: enlarging the tangent space can only reduce the off-manifold residual. The interesting content is in the *size* of the reduction, which quantifies how much of the one-field diagnostic's signal was actually chemical-potential content that the two-field manifold absorbs.

E. Pythagorean decomposition

The decomposition (22) already contains the Pythagorean identity; we state it as a named result because Paper II (Sec. V E therein) relies on it for the upgrade-decision comparison.

Proposition 5 (Pythagorean decomposition, all ξ)**.** *The one-field diagnostic decomposes exactly into an η-tangent component and a genuine spectral-distortion residual:*

$$\big[\mathcal{D}^{(1\mathrm{D})}_{\geq 2}\big]^2 = \big\|\Pi_{V_2\ominus V_1} G_\xi\big\|_*^2 + \big[\mathcal{D}^{(2\mathrm{D})}_{\geq 2}\big]^2. \quad (23)$$

The first term is the η-tangent component (absorbed by the two-field manifold); the second is the genuine spectral distortion.

Proof. The proof proceeds in three steps.

Step 1: orthogonal decomposition of $V_1^\perp$. The Hilbert space $L^2(w_*)$ admits the orthogonal direct-sum decomposition

$$L^2(w_*) = V_1 \oplus (V_2 \ominus V_1) \oplus V_2^\perp, \quad (24)$$

where $V_2 \ominus V_1 := V_2 \cap V_1^\perp$ is the one-dimensional subspace of V_2 orthogonal to V_1. This subspace is spanned by $\tilde{v}_2 := v_2 - \Pi_1 v_2$—the component of $v_2 = 1$ orthogonal to $v_1 = x/\Theta^2$—and is nonempty because v_1 and v_2 are linearly independent (Proposition 2, Paper II). The orthogonal complement of V_1 decomposes as

$$V_1^\perp = (V_2 \ominus V_1) \oplus V_2^\perp. \quad (25)$$

Step 2: decompose the 1D residual. The 1D diagnostic residual $G_\xi - \Pi_1 G_\xi$ lies in $V_1^\perp$. Applying the decomposition (25):

$$G_\xi - \Pi_1 G_\xi = \Pi_{V_2\ominus V_1} G_\xi + \Pi_{V_2^\perp} G_\xi. \quad (26)$$

The two terms are orthogonal by construction.

Step 3: Pythagorean theorem. Taking the squared norm of both sides of (26) and using orthogonality:

$$\|G_\xi - \Pi_1 G_\xi\|_*^2 = \|\Pi_{V_2\ominus V_1} G_\xi\|_*^2 + \|\Pi_{V_2^\perp} G_\xi\|_*^2.$$

The left side is $[\mathcal{D}^{(1\mathrm{D})}_{\geq 2}]^2$ by definition (20). The second term on the right is $[\mathcal{D}^{(2\mathrm{D})}_{\geq 2}]^2$ because $\Pi_{V_2^\perp} G_\xi = G_\xi - \Pi_2 G_\xi$ (the full residual after projecting onto V_2). □

Physical content. The first term on the right of (23)—the η-tangent component—is the part of the collision source that the one-field manifold cannot absorb because it lacks the η-direction, but that the two-field manifold absorbs exactly. It is the "false trigger" of the one-field diagnostic: the 1D diagnostic reports this component as off-manifold spectral distortion, when in reality it is chemical-potential content. The second term is the genuine spectral distortion that neither manifold can absorb.

The η-tangent fraction. We define

$$\mathcal{F}_\eta := \frac{\|\Pi_{V_2\ominus V_1} G_\xi\|_*^2}{[\mathcal{D}^{(1\mathrm{D})}_{\geq 2}]^2} = 1 - \frac{[\mathcal{D}^{(2\mathrm{D})}_{\geq 2}]^2}{[\mathcal{D}^{(1\mathrm{D})}_{\geq 2}]^2} \quad (27)$$

as the fraction of the one-field diagnostic's signal that is η-tangent content. At moderate FD degeneracy ($\eta_0 \approx 3$), this fraction is $\mathcal{F}_\eta = 0.985$: 98.5% of the one-field signal is false-trigger content (Paper II, Table X). At non-degenerate conditions (MB or $\eta_0 = 0$ FD), $\mathcal{F}_\eta \approx 0.85$: even in the classical limit, the one-field diagnostic over-estimates by a factor of ~ 7 (Fig. 2).

F. Principal angle between tangent spaces

The η-tangent fraction $\mathcal{F}_\eta$ depends on the geometry of the collision source relative to the tangent spaces. A collision-source-independent quantity that characterises the manifold geometry alone is the principal angle between V_1 and the η-direction.

Proposition 6 (Principal angle, all ξ)**.** *The angle θ_ξ between the temperature tangent direction $v_1 = x/\Theta^2$ and the chemical-potential tangent direction $v_2 = 1$ in $\langle\cdot,\cdot\rangle_*$ is*

$$\cos\theta_\xi = \frac{[\mathbf{G}_\xi]_{12}}{\sqrt{[\mathbf{G}_\xi]_{11}\,[\mathbf{G}_\xi]_{22}}} = \frac{\langle v_1, v_2\rangle_*}{\|v_1\|_*\,\|v_2\|_*}. \quad (28)$$

The angle controls but does not uniquely determine the Gram-matrix condition number: for a general 2×2 positive-definite matrix,

$$\kappa(\mathbf{G}_\xi) = \frac{G_{11}+G_{22} + \sqrt{(G_{11}-G_{22})^2 + 4G_{12}^2}}{G_{11}+G_{22} - \sqrt{(G_{11}-G_{22})^2 + 4G_{12}^2}}, \quad (29)$$

which reduces to $(1+|\cos\theta|)/(1-|\cos\theta|)$ only when $\|v_1\|_* = \|v_2\|_*$. In the physical configurations of interest, the norm ratio $\|v_1\|_*/\|v_2\|_*$ differs from unity, so the full formula (29) must be used.

Proof. Equation (28) is the definition of the angle between two vectors in an inner-product space. For a 2×2 positive-definite matrix with entries G_{11}, G_{22}, G_{12}, the eigenvalues are $\lambda_\pm = \frac{1}{2}(G_{11}+G_{22}) \pm \frac{1}{2}\sqrt{(G_{11}-G_{22})^2 + 4G_{12}^2}$, giving (29). □

Statistics dependence (Fig. 1). For MB at $\Theta = 1.2$: $\theta \approx 26.6°$, $\kappa \approx 78$. The angle is η_0-independent because the e^{η_0} factor cancels. For FD at $\eta_0 = 10$: $\theta \approx 9.5°$, $\kappa \approx 3207$. The angle shrinks because the Fermi-surface concentration makes $v_1 \approx \eta_0/\Theta$ approximately constant over the narrow support of w_*, so v_1 and $v_2 = 1$ become nearly collinear. For BE at $\eta_0 = -0.3$: $\theta \approx 28.5°$, $\kappa \approx 63$. Bose enhancement broadens the support of w_* and separates the two tangent directions more effectively than in the classical case.

G. Leading normal mode and upgrade prescription

The diagnostic residual $G_\xi - \Pi_2 G_\xi \in V_2^\perp$ is an energy-dependent function that can be expanded in a generalised Laguerre basis $\{L_n^{(\alpha_\xi)}(x/\Theta)\}_{n\geq 0}$ adapted to the weight w_*. The leading mode $n = 2$ (quadratic in x) dominates the residual whenever the collision source is smooth in energy.

The leading-mode concentration $|c_2|^2/[\mathcal{D}_{\geq 2}^{(2D)}]^2$—the fraction of the *two-field* diagnostic captured by a single energy-shape mode—determines the upgrade prescription: if $|c_2|^2/[\mathcal{D}_{\geq 2}^{(2D)}]^2 > 0.8$, a one-mode correction (adding a single energy-shape degree of freedom) suffices; if the concentration is low, the spectral distortion is broadly distributed and a multigroup energy discretisation is the appropriate response.

For BGK collisions (Table II), the distinction is dramatic. At MB and BE, the 2D leading-mode concentration exceeds 89%: a one-mode correction suffices. At FD $\eta_0 = 3$, the 2D concentration drops to 13%: the genuine spectral distortion is broadly distributed, requiring multigroup treatment. The one-field diagnostic would *incorrectly* prescribe one-mode correction (81% 1D concentration) because the dominant η-tangent component—which the one-field manifold misclassifies as $n = 2$ spectral content—projects strongly onto the leading Laguerre mode. The two-field diagnostic, by filtering out the η-tangent content via the Pythagorean decomposition (Proposition 5), reveals the true spectral structure of the residual and provides the correct upgrade prescription.

This distinction is the operational payoff of the Pythagorean decomposition: not merely a smaller diagnostic value, but a qualitatively different upgrade decision.

H. Worked example: BGK collision at moderate FD degeneracy

We illustrate the full diagnostic chain at $(\Theta, \eta_0) = (1.2, 3)$ with FD statistics and a BGK isotropisation collision $C = (f_{\rm iso} - f)/\tau$.

The entropy-normalised collision source $G_\xi(x)$ is computed by evaluating $C[f]$ at the current state and dividing by $f(1-f)$. Projecting onto $V_1 = \mathrm{span}\{x/\Theta^2\}$ gives the one-field tangent component; projecting onto $V_2 = \mathrm{span}\{x/\Theta^2, 1\}$ gives the two-field tangent component.

The results (Table II, FD $\eta_0 = 3$ row): $\mathcal{D}_{\geq 2}^{(1D)} = 0.226$ (relative to $\|G_\xi\|_*$) and $\mathcal{D}_{\geq 2}^{(2D)} = 0.037$. The reduction factor is 6.1×: the two-field diagnostic is six times smaller than the one-field diagnostic. The η-tangent fraction is $\mathcal{F}_\eta = 1 - (0.037/0.226)^2 = 0.97$: 97% of the one-field signal is chemical-potential content that the two-field manifold absorbs.

The leading-mode concentrations differ qualitatively: 81% on the 1D residual (suggesting one-mode correction) versus 13% on the 2D residual (indicating multigroup discretisation). The one-field diagnostic would make the wrong upgrade decision at this configuration.

I. Diagnostic survey across statistics

Table II reports the diagnostic values for BGK collisions at representative configurations across all three statistics. The reduction factor $\mathcal{D}_{\geq 2}^{(1D)}/\mathcal{D}_{\geq 2}^{(2D)}$ ranges from 2.4× (BE at $\eta_0 = -0.5$) to 6.1× (FD at $\eta_0 = 3$).

The reduction is largest at moderate FD degeneracy, where the BGK source has a substantial η-direction component: the isotropisation operator shifts both Θ and η toward equilibrium, and the η-shift is large relative to the spectral distortion because Pauli blocking limits the energy-shape response. At low degeneracy (MB or $\eta_0 = 0$ FD) and at deep BE bulk ($\eta_0 = -3$), the η-component is smaller relative to the total source, and the reduction is more modest ($\sim 2.6\times$).

Multipole relaxation gives $\mathcal{D}_{\geq 2}^{(2D)} < 10^{-12}$ at all tested configurations—effectively zero to quadrature precision. This is the expected behaviour from Theorem 3: a multipole-relaxation operator lies in V_2 by construction.

The practical implication for transport codes: the one-field diagnostic of Paper I *overestimates* the off-manifold content for degenerate systems. A transport code using the 1D diagnostic would trigger unnecessary upgrades or multigroup switches in configurations where the two-field manifold is adequate. The 2D diagnostic, by absorbing the η-tangent content, provides both a smaller (more accurate) measure of the genuine spectral distortion and a qualitatively correct upgrade prescription.

J. Triple role of the Gram matrix

The tangency diagnostic and the entropy concavity of §II are dual perspectives on the same geometry. Theorem 1 says that the Gram matrix $\mathbf{G}_\xi$ measures the entropy curvature along tangent directions. Theorem 3 says that the same Gram matrix—through the projection formula (19)—determines which collision sources are tangent and which produce genuine off-manifold content. The Gram matrix simultaneously serves three distinct roles:

Identifiability metric (Paper II): $\mathbf{G}_\xi$ is the Fisher-type information matrix. Its positive definiteness ensures that Θ and η can be separately identified from moment data.

Entropy Hessian (Theorem 1): $\mathbf{G}_\xi$ is the negative second variation of the entropy. Its positive definiteness ensures strict entropy concavity on the manifold.

Tangency projection kernel (Theorem 3): $\mathbf{G}_\xi$ determines the projection Π_2 that splits the collision source into tangent and off-manifold components. Its condition number determines the η-tangent fraction $\mathcal{F}_\eta$.

This triple identification is not a coincidence; it reflects the exponential-family structure of the distribution (1). For any exponential family, the Fisher information, the entropy Hessian, and the natural-parameter Gram matrix coincide (Amari [5], Chapter 3). The (Θ, η) manifold is a two-parameter exponential family with natural parameters $\boldsymbol{\theta} = (-1/\Theta, \eta)$ and sufficient statistics $\boldsymbol{t} = (x, 1)$. Theorems 1 and 3, and the Pythagorean decomposition (Proposition 5), are formal consequences of this structure applied to the specific inner product (2).

IV. THERMODYNAMIC CONVEXITY GROUNDWORK

A. Motivation: Godunov-symmetric structure

The Godunov formulation of moment-based kinetic theory requires a strictly convex thermodynamic potential whose Hessian serves as the symmetrising matrix for the flux Jacobian [3, 4]. Symmetric hyperbolicity of the resulting PDE system guarantees finite-speed signal propagation and a local well-posedness theory; its absence permits nonphysical instabilities. On the one-field Teff manifold, Paper I established the ray-wise convexity and entropy structure relevant to a future Godunov formulation—identifying a convex potential $\Psi(a) = -\int Q(\Theta(\mu))\,\mathrm{d}\mu$ whose ray-wise Hessian is positive definite because $W'(\Theta) > 0$ for all $\Theta > 0$—but did not prove symmetric hyperbolicity of the projected PDE hierarchy.

On the extended (Θ, η) manifold, the ray-wise potential acquires an additional direction. The necessary condition for the Godunov structure to survive is that the 2×2 ray-wise Hessian—the generalisation of the scalar $W'(\Theta)$ to the two-parameter family—be positive definite. This section establishes this positivity and shows that the proof is a direct consequence of the same tangent-vector non-proportionality that certifies identifiability (Paper II, Proposition 2) and entropy concavity (Theorem 1).

B. Ray-wise thermodynamic potential

The ray-wise *convex* potential for the two-field manifold is

$$\Psi_\xi^{(p)}(\Theta, \eta) = -\int_0^\infty \sigma_\xi\big(f(x; \Theta, \eta)\big)\, x^p\, \mathrm{d}x, \tag{30}$$

where σ_ξ is the entropy generating function (3) and p is the moment weight: $p = 3$ for the energy-density sector (governing the evolution of $\mathcal{E}_0, \mathcal{F}_1, \mathcal{P}_2$) and $p = 2$ for the number-density sector (governing $\mathcal{N}_0, \mathcal{N}_1$). The minus sign makes Ψ convex (the entropy $\int \sigma_\xi$ is concave; negating it produces the convex thermodynamic potential needed for the Godunov symmetrisation).

The 2×2 Hessian of $\Psi_\xi^{(p)}$ with respect to the per-ray parameters $\boldsymbol{\alpha} = (\Theta, \eta)$ is

$$[H_\xi^{(p)}]_{ij} = \frac{\partial^2 \Psi_\xi^{(p)}}{\partial \alpha_i \partial \alpha_j} = \int_0^\infty f(1{+}\xi f)\, v_i(x)\, v_j(x)\, x^p\, \mathrm{d}x, \tag{31}$$

where $v_1 = x/\Theta^2$ and $v_2 = 1$ are the tangent-direction functions (the same as in §II C). The derivation follows the same three-step calculation as for the Gram–Hessian identity (Theorem 1), with x^3 replaced by x^p. The Hessian (31) is the Gram matrix of $\{v_1, v_2\}$ in the p-weighted inner product $\langle g, h\rangle_p := \int g\, h\, f(1{+}\xi f)\, x^p\, \mathrm{d}x$.

C. Positivity

Proposition 7 (Ray-wise thermodynamic Hessian positivity, all ξ)**.** *For all $\Theta > 0$, admissible η, and $\xi \in \{0, -1, +1\}$, the 2×2 Hessian (31) is positive definite for $p = 2$ (number-density weighting) and $p = 3$ (energy-density weighting).*

Proof. The weight $w_p = f(1{+}\xi f)\, x^p$ is strictly positive for all $x > 0$: we have $f > 0$, $1{+}\xi f > 0$ on the physical domain (for FD, $0 < f < 1$ so $1{-}f > 0$; for BE, $f > 0$ and $1{+}f > 0$; for MB, $f > 0$), and $x^p > 0$ for $p > 0$. Hence $\langle\cdot,\cdot\rangle_p$ is a genuine inner product on the space of measurable functions square-integrable against w_p.

To show positive definiteness of $H_\xi^{(p)}$, we verify that $\det H_\xi^{(p)} > 0$ and $[H_\xi^{(p)}]_{11} > 0$. The diagonal entry $[H_\xi^{(p)}]_{11} = \langle v_1, v_1\rangle_p = \int f(1{+}\xi f)(x/\Theta^2)^2 x^p\, \mathrm{d}x > 0$ because the integrand is strictly positive. The determinant is the Gram determinant:

$$\det H_\xi^{(p)} = \langle v_1, v_1\rangle_p\, \langle v_2, v_2\rangle_p - \langle v_1, v_2\rangle_p^2. \tag{32}$$

By the Cauchy–Schwarz inequality in $L^2(w_p)$, $\det H_\xi^{(p)} = 0$ if and only if v_1 and v_2 are proportional w_p-almost everywhere. Since $v_1(x) = x/\Theta^2$ is strictly increasing in x while $v_2(x) = 1$ is constant, they are not proportional on any interval. Hence $\det H_\xi^{(p)} > 0$, and combined with $[H_\xi^{(p)}]_{11} > 0$ this gives positive definiteness. □

Remark (generality). The proof holds for any $p > 0$; we restrict the statement to $p = 2$ and $p = 3$ because these are the physically relevant moment weights. The argument is identical in structure to the proof of tangent-space independence (Paper II, Proposition 2) and the Gram–Hessian identity (Theorem 1): all three rely on the non-proportionality of v_1 and v_2 in a positive-definite inner product. This structural unity is a further consequence of the exponential-family property.

D. Numerical verification and p-dependence

Table III reports the Hessian eigenvalues and condition numbers for $p = 2$ (number sector) and $p = 3$ (energy sector) across representative configurations. Two patterns emerge.

First, the $p = 2$ Hessian is systematically better conditioned than the $p = 3$ Hessian at the same (Θ, η_0, ξ). At FD $(\Theta, \eta_0) = (1.0, 6)$: $\kappa_2 \approx 898$ while $\kappa_3 \approx 1045$—the number-sector Hessian is 14% better conditioned. The physical origin is the energy weighting: the x^2 weight in w_2 suppresses the high-energy tail relative to the x^3 weight in w_3. Since the tangent vectors $v_1 = x/\Theta^2$ and $v_2 = 1$ are most collinear at high x (where v_1 is large and varies slowly relative to v_2), reducing the high-energy contribution improves the angular separation and hence the conditioning.

Second, at MB both κ_2 and κ_3 are η_0-independent, consistent with the self-reproduction property (Paper II, Proposition 16). For BE at $\eta_0 = -1$, both Hessians are well-conditioned ($\kappa < 60$).

E. Full angular Hessian positivity

The full angular thermodynamic potential is

$$\Psi(\boldsymbol{\alpha}) = \int_{-1}^{1} \Psi_\xi^{(p)}\big(\Theta(\mu), \eta(\mu)\big)\,\mathrm{d}\mu, \tag{33}$$

where $\Theta(\mu)$ and $\eta(\mu)$ are given by the $(L_\Theta = 2,\ L_\eta = 1)$ parameterisation and $\boldsymbol{\alpha} = (\Theta_0, A_\Theta, Q_\Theta, \eta_0, A_\eta)$. The ray-wise Hessian $H_\xi^{(p)}(\Theta(\mu), \eta(\mu))$ is the 2×2 positive-definite matrix of Proposition 7. We now show that the full 5×5 parameter Hessian $\nabla^2_{\boldsymbol{\alpha}}\Psi$ is also positive definite.

Proposition 8 (Full angular Hessian positivity)**.** *For the $(L_\Theta = 2,\ L_\eta = 1)$ parameterisation and each energy weight $p \in \{2, 3\}$, the 5×5 Hessian $\nabla^2_{\boldsymbol{\alpha}}\Psi^{(p)}$ is positive definite at all admissible $\boldsymbol{\alpha}$.*

Proof. The proof uses the *additive* angular parameterisation, in which $\boldsymbol{\alpha} = (\Theta_0, A_\Theta, Q_\Theta, \eta_0, A_\eta)$ enters linearly:

$$\begin{aligned}\Theta(\mu) &= \Theta_0 + A_\Theta\,\mu + Q_\Theta(\mu^2 - \tfrac{1}{3}),\\ \eta(\mu) &= \eta_0 + A_\eta\,\mu.\end{aligned} \tag{34}$$

(Paper II uses the multiplicative form $\Theta(\mu) = \Theta_0(1 + \hat{A}\mu + \hat{Q}P_2)$; the additive coordinates are related by $A_\Theta = \Theta_0\hat{A}$, $Q_\Theta = \Theta_0\hat{Q}$. Positive definiteness is a coordinate-independent property of the quadratic form, so the result holds in either parameterisation.) Define the per-ray parameter vector $z(\mu) := (\Theta(\mu), \eta(\mu)) \in \mathbb{R}^2$ and the 2×5 matrix $B(\mu)$ such that $z(\mu) = B(\mu)\,\boldsymbol{\alpha}$. Linearity means $\partial^2 z/\partial\alpha_k\partial\alpha_l = 0$, so the second variation of Ψ contains no chain-rule correction and reduces to

$$\delta^2\Psi[\delta\boldsymbol{\alpha}] = \int_{-1}^{1} \delta z(\mu)^T\, H_\xi^{(p)}\big(z(\mu)\big) \times \delta z(\mu)\,\mathrm{d}\mu, \tag{35}$$

where $\delta z(\mu) = B(\mu)\,\delta\boldsymbol{\alpha}$. Since $H_\xi^{(p)}$ is positive definite at every μ (Proposition 7), the integrand is ≥ 0, with equality at a given μ only if $\delta z(\mu) = 0$.

Suppose $\delta^2\Psi[\delta\boldsymbol{\alpha}] = 0$. Then $\delta z(\mu) = 0$ for almost every μ, i.e.

$$\delta\Theta(\mu) = 0, \qquad \delta\eta(\mu) = 0 \qquad \text{a.e.}$$

Since $\delta\Theta(\mu) = \delta\Theta_0 + \delta A_\Theta\,\mu + \delta Q_\Theta(\mu^2 - 1/3)$ is a polynomial, vanishing a.e. implies vanishing identically, so $\delta\Theta_0 = \delta A_\Theta = \delta Q_\Theta = 0$. Similarly $\delta\eta(\mu) = \delta\eta_0 + \delta A_\eta\,\mu = 0$ identically gives $\delta\eta_0 = \delta A_\eta = 0$. Hence $\delta\boldsymbol{\alpha} = 0$.

Therefore $\delta^2\Psi[\delta\boldsymbol{\alpha}] > 0$ for all $\delta\boldsymbol{\alpha} \neq \mathbf{0}$, establishing positive definiteness. ◻

Remark 9 (Scope: thermodynamic Hessian versus symmetric hyperbolicity). Proposition 8 establishes that the thermodynamic potential Ψ is strictly convex in the five-parameter state vector. This is a *necessary* ingredient for the Godunov symmetric formulation, but not by itself sufficient: symmetric hyperbolicity of the projected PDE system requires additionally that the Hessian symmetrise the flux Jacobian, which involves the spatial-transport structure (characteristic speeds, numerical flux) not addressed in the present paper. The remaining gap is the flux-Jacobian coupling, not the thermodynamic convexity.

V. KNOWN-LIMIT REINTERPRETATION

The five known limits verified numerically in Paper II (Sec. IV therein) are now re-examined through the entropy and tangency lens. Each limit is a consistency check of the geometric structures established in §§II–IV; the PASS/CHAR status from Paper II is unchanged, but the geometric interpretation is new.

A. L1: Non-degenerate limit ($\eta_0 \to -\infty$)

As $\eta_0 \to -\infty$, both FD and BE distributions converge to Maxwell–Boltzmann. The inner-product weight $w_* = f(1+\xi f)\,x^3$ approaches the Boltzmann weight $f\,x^3$

because the quantum factor $(1+\xi f)$ tends to unity when $f \ll 1$.

Entropy geometry. The Gram matrix $\mathbf{G}_\xi$ converges to its MB value. By Theorem 1, so does the entropy Hessian. The entropy surface at deep non-degeneracy has the universal MB shape: $\kappa \to \kappa_{\rm MB} \approx 78$, with the principal angle $\theta \to 26.6^\circ$ (Proposition 6). Numerically, the FD and BE Gram eigenvalues agree with MB to $< 10^{-15}$ at $\eta_0 = -12$ (Paper II, Table IX).

Tangency. The 1D and 2D diagnostics both converge to their MB values. The η-tangent fraction $\mathcal{F}_\eta$ (27) converges to $\mathcal{F}_\eta^{\rm MB} \approx 0.85$ for BGK collisions. The monotone reduction (Proposition 4) is preserved uniformly in the classical limit.

B. L2: Isotropic limit ($A_\Theta = A_\eta = 0$)

At homogeneous equilibrium the per-ray entropy is maximised: $\delta S_\xi = 0$ (first variation vanishes) and $\delta^2 S_\xi < 0$ (Corollary 2). The collision operator vanishes ($C[f_{\rm eq}] = 0$), so the entropy-normalised source $G_\xi = 0$ and both diagnostics vanish identically (Theorem 3).

The Gram matrix at isotropic $\Theta = \Theta_0$ gives the equilibrium entropy curvature; its eigenvalues depend only on (Θ_0, η_0, ξ). The concavity (Corollary 2) ensures that any tangent perturbation away from equilibrium decreases the entropy—this is the local manifestation of the H-theorem on the Teff manifold.

C. L3: Scalar-η reduction ($A_\eta = 0$)

At vanishing chemical-potential dipole, the η-direction is present in the tangent space but carries no angular anisotropy: the angular structure is determined entirely by $\Theta(\mu)$.

Entropy geometry. The scalar η_0 shifts both Gram eigenvalues by the same multiplicative factor for MB (the common factor e^{η_0} cancels in every ratio; Paper II, Proposition 16) but by different factors for FD and BE. The entropy curvature along η remains strictly positive (Corollary 2): scalar η_0 does not flatten the entropy surface. The cross-sector invariance of $f_\mathcal{N}/f_\mathcal{E}$ (Paper II, Proposition 17) is the observable manifestation: a direction-independent chemical potential moves the distribution uniformly along the soft eigenvector, producing no angular signature in the flux ratio.

Tangency. At $A_\eta = 0$, the BGK collision source still has a nonzero η-component (because isotropisation changes η as well as Θ). The Pythagorean decomposition (Proposition 5) applies, but the η-tangent fraction is set by the collision geometry at the specific (Θ, η_0) rather than by the angular structure.

D. L4: Pure η-dipole ($A_\Theta = 0$, $A_\eta \neq 0$)

All anisotropy arises from direction-dependent degeneracy. On the one-field manifold, the tangency diagnostic interprets *all* anisotropy as off-manifold content. The Pythagorean decomposition quantifies: $> 96\%$ of the one-field signal is η-tangent content that the two-field manifold absorbs exactly. The 2D diagnostic reports only the residual spectral distortion, which is $< 8\%$ of the source norm at moderate degeneracy.

This is the most physically significant finding of the tangency analysis. For degenerate systems with number-conserving collisions (e.g. neutrino charged-current processes in core-collapse supernovae), the one-field diagnostic would trigger unnecessary model upgrades in configurations where the two-field manifold is adequate. The overestimation is largest at moderate degeneracy ($\eta_0 \sim 3$: $6.1\times$) and decreases at both low and very high degeneracy.

E. L5: Conditioning characterisation

The condition number $\kappa(\mathbf{G}_\xi)$ is the ratio of entropy curvatures along the two tangent directions (Proposition 6, Eq. (29)). Three regimes:

At FD $\eta_0 = 6$: $\kappa \approx 1045$, $\theta \approx 14^\circ$. The η-direction has 0.1% of the entropy curvature of the Θ-direction. Perturbations in η change the distribution very little because Pauli blocking limits the occupation to $f \lesssim 1$ over most of the spectrum.

At FD $\eta_0 = 10$: $\kappa \approx 3207$, $\theta \approx 9.5^\circ$. Entropy-based inversion methods will converge slowly along the η-direction.

At BE $\eta_0 = -0.3$: $\kappa \approx 63$, $\theta \approx 28.5^\circ$. Bose enhancement improves the curvature separation.

In all cases both eigenvalues remain strictly positive (Corollary 2); the conditioning issue is quantitative, not qualitative. The Jacobian remains full-rank (Paper II, Table II), confirming that high conditioning is a practical caveat for numerical inversion, not a structural obstruction to the two-field representation.

VI. DISCUSSION

A. Exponential-family origin and e-projection

The distribution (1) is a two-parameter exponential family with natural parameters $\boldsymbol{\theta} = (-1/\Theta,\, \eta)$ and sufficient statistics $\boldsymbol{t} = (x,\, 1)$:

$$f = \Phi_\xi(\boldsymbol{\theta} \cdot \boldsymbol{t}) = [\exp(-\boldsymbol{\theta} \cdot \boldsymbol{t}) - \xi]^{-1}. \qquad (36)$$

For any exponential family, three fundamental objects coincide [5]: the Fisher information matrix $\mathcal{I}_{ij} = -\mathbb{E}[\partial_i \partial_j \ln f]$, the (negative) entropy Hessian $-\partial_i \partial_j S_\xi$, and the Gram matrix $\langle \partial_i \ln f,\, \partial_j \ln f \rangle_{w_*}$. This triple identification explains Theorem 1 (entropy = Gram), Theorem 3 (tangency via Gram projection), and the Pythagorean decomposition (Proposition 5) at a unified structural level.

The exponential-family structure also implies that the *e-projection* (exponential-family maximum-likelihood projection) of any distribution g onto the manifold $\mathcal{M}$ minimises the Kullback–Leibler divergence $D_{\mathrm{KL}}(g\|f)$ over $f \in \mathcal{M}$, and that the resulting projection is the unique distribution in $\mathcal{M}$ matching the expected sufficient statistics $\mathbb{E}_g[\boldsymbol{t}]$ [6]. The tangency diagnostic (Theorem 3) detects departure from this e-projection: $\mathcal{D}^{(2\mathrm{D})}_{\geq 2} > 0$ means the collision source drives the distribution away from the e-projection's tangent space.

The Pythagorean theorem of information geometry [5, 6] gives the global decomposition:

$$D_{\mathrm{KL}}(g\|f_{\mathrm{ref}}) = D_{\mathrm{KL}}(g\|f_{\mathrm{proj}}) + D_{\mathrm{KL}}(f_{\mathrm{proj}}\|f_{\mathrm{ref}}), \quad (37)$$

where f_{proj} is the e-projection of g onto the manifold. The local Pythagorean decomposition (Proposition 5) is the infinitesimal version of (37), applied to the nested manifolds $\mathcal{M}_{\Theta} \subset \mathcal{M}_{\Theta,\eta}$. The first term in (23) corresponds to $D_{\mathrm{KL}}(g\|f^{(1\mathrm{D})}_{\mathrm{proj}}) - D_{\mathrm{KL}}(g\|f^{(2\mathrm{D})}_{\mathrm{proj}})$—the KL-divergence reduction from enlarging the manifold—while the second term is the residual divergence from the two-field manifold.

B. Entropy-projection inequality: status

The entropy-projection inequality of Paper I (Theorem 9 therein) states that the entropy of the manifold-projected distribution is at least as large as that of any distribution sharing the same moments. Its proof requires global convexity of the entropy functional on the full distribution space, not merely local concavity on the tangent space.

Corollary 2 establishes the local result for the two-field manifold. The global result—extending the entropy-projection inequality from the one-field to the two-field manifold—would follow from standard information-geometric arguments if the per-ray moment map (mapping distributions to their sufficient statistics) is surjective onto the interior of the moment polytope [6]. We have not completed this verification for the two-field case and defer it to future work. The local entropy concavity established here is the structurally essential result; the global projection inequality, if it holds, is a formal upgrade that does not affect the tangency diagnostic or the Godunov structure.

C. Summary of results

Established (all for $\xi \in \{0, -1, +1\}$):

(i) Gram–Hessian identity: the entropy Hessian restricted to the (Θ, η) tangent space is identically the Gram matrix (Theorem 1). Strict entropy concavity follows (Corollary 2).

(ii) Tangency diagnostic survival: the diagnostic vanishes iff the collision source is tangent to the two-field manifold (Theorem 3). The monotone reduction (Proposition 4) and Pythagorean decomposition (Proposition 5) quantify the relationship to the one-field diagnostic.

(iii) Principal angle characterisation (Proposition 6): the condition number is controlled by the angle between the temperature and chemical-potential tangent directions together with their norm ratio.

(iv) Ray-wise thermodynamic Hessian positivity for $p = 2$ and $p = 3$ (Proposition 7): the necessary building block for Godunov-symmetric structure.

(v) Triple role of the Gram matrix: identifiability metric, entropy Hessian, and tangency projection kernel—unified by the exponential-family structure (§VI A).

(vi) Full 5×5 angular Hessian positive definiteness for the ($L_{\Theta} = 2$, $L_{\eta} = 1$) parameterisation (Proposition 8): the thermodynamic convexity groundwork for the Godunov formulation is complete.

Not established:

(i) Global entropy-projection inequality for the two-field manifold (§VI B).

(ii) Symmetric hyperbolicity of the (Θ, η) PDE system: the flux-Jacobian symmetrisation by the thermodynamic Hessian remains to be verified (Remark 9).

(iii) Transport-performance comparison or astrophysical benchmarks.

The contributions are structural: they establish the geometric and thermodynamic foundations needed for a future transport implementation, without providing that implementation.

D. Outlook

Paper III completes the mathematical foundations of the Teff two-field manifold: identifiability (Paper II), entropy geometry (this paper), and tangency diagnostic survival. The Gram matrix introduced in Paper II as an identifiability metric is shown here to be simultaneously the tangent-space entropy Hessian and the projection kernel governing the two-field tangency diagnostic. Three developments follow.

First, Proposition 8 establishes the thermodynamic convexity of Ψ in the five-parameter state vector. The next step toward symmetric hyperbolicity is verifying that this Hessian symmetrises the flux Jacobian of the projected transport system—a computation involving the characteristic speeds and spatial discretisation that lies outside the present scope.

Second, the diagnostic reduction (Proposition 5, Table II) should be tested for collision operators beyond BGK and multipole relaxation—in particular, for neutrino–matter interactions (charged-current absorption/emission, neutral-current scattering) and Compton scattering, where the energy-exchange physics is richer

TABLE I. Gram matrix eigenvalues, condition number, and principal angle at $\Theta = 1.2$ for all three statistics. MB is η_0-independent; FD degrades with degeneracy; BE improves near the condensation edge.

ξ	η_0	$\lambda_{\min}$	$\lambda_{\max}$	κ	θ [°]	η-align.
MB	any	0.27	21	78	26.6	0.972
FD	0	0.31	29	92	25.2	0.973
	3	0.99	350	355	19.7	0.980
	6	4.1	6600	1604	14.0	0.989
	10	—	—	3207	9.5	0.994
BE	-0.3	2.9	170	63	28.5	0.970
	-1	0.96	68	71	27.3	0.971
	-5	0.016	1.2	78	26.7	0.972

TABLE II. Tangency diagnostic on the 1D (Θ-only) and 2D (Θ, η) tangent spaces for BGK collisions. All values relative to $\|G_\xi\|_*$. Proposition 4 ($\mathcal{D}^{(2D)}_{\geq 2} \leq \mathcal{D}^{(1D)}_{\geq 2}$) satisfied in every case. Last row: multipole relaxation (tangent by construction).

ξ	η_0	Θ	$D^{(1D)}$	$D^{(2D)}$	Redn.	η-frac.
MB	0	1.2	0.186	0.071	2.6×	85%
FD	0	1.2	0.186	0.071	2.6×	85%
FD	3	1.2	0.226	0.037	6.1×	97%
FD	6	1.2	0.320	0.072	4.5×	95%
BE	-0.5	1.2	0.195	0.082	2.4×	82%
BE	-3	1.2	0.187	0.072	2.6×	85%
all	—	—	$< 10^{-12}$	$< 10^{-12}$	—	—

than the BGK model. The η-tangent fraction $\mathcal{F}_\eta$ may differ significantly for these operators.

The present paper does not establish a global entropy-projection inequality, symmetric hyperbolicity of the PDE system (the flux-Jacobian coupling remains open), or transport-performance superiority. These remain deferred results.

Third, coupling the five-parameter state vector to a transport solver with an explicit lepton-number evolution equation would test whether the structural non-redundancy and diagnostic improvements translate to practical closure advantages. Paper IV addresses the observable semantics and reconstructive adequacy of such a solver; Paper V establishes the intensity-side bridge to polarised radiative transfer.

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TABLE III. Ray-wise thermodynamic Hessian (Proposition 7) for $p = 2$ (number) and $p = 3$ (energy). All entries are positive definite. The $p = 2$ Hessian is systematically better conditioned.

ξ	Θ	η_0	p	$\lambda_{\min}$	κ
MB	1.0	0	2	0.34	71
	1.0	0	3	0.94	128
FD	1.0	3	2	1.5	216
	1.0	3	3	6.3	297
FD	1.0	6	2	2.3	898
	1.0	6	3	15.8	1045
BE	1.0	-1	2	0.25	42
	1.0	-1	3	0.71	56

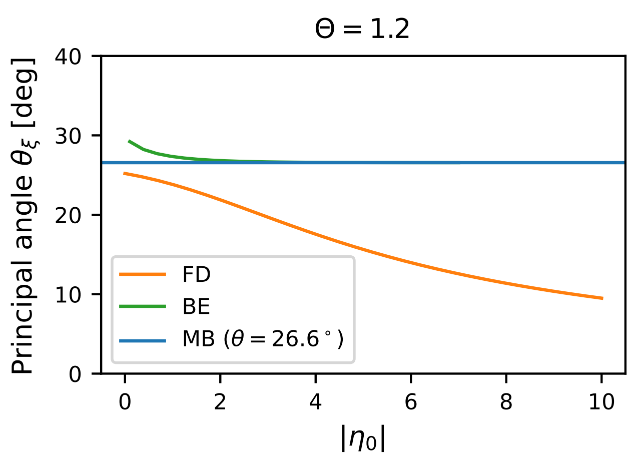

FIG. 1. Principal angle θ_ξ between $v_1 = x/\Theta^2$ and $v_2 = 1$ in $\langle\cdot,\cdot\rangle_*$ versus $|\eta_0|$ at $\Theta = 1.2$. MB (blue): constant at 26.6°. FD (orange): decreases monotonically due to Pauli blocking. BE (green): exceeds θ_{MB} throughout the admissible domain. The angle governs the conditioning (Proposition 6).

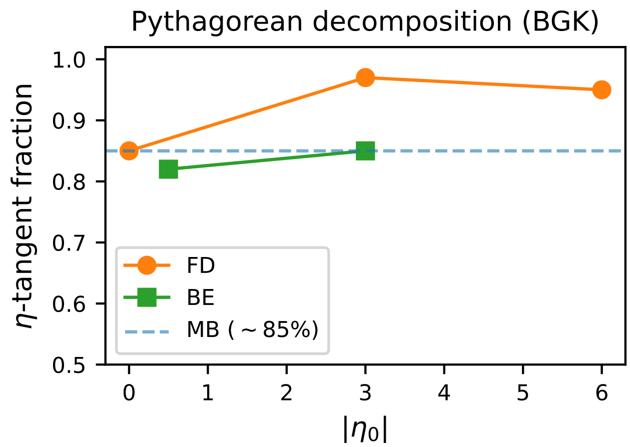

FIG. 2. Pythagorean decomposition (Proposition 5) for BGK collisions at $\Theta = 1.2$. Solid: η-tangent fraction $\|\Pi_{V_2 \ominus V_1} G_\xi\|_*^2 / [\mathcal{D}_{\geq 2}^{(1D)}]^2$. FD at moderate degeneracy ($\eta_0 \approx 3$): 97% of the 1D signal is η-tangent. MB and deep-bulk BE: $\sim 85\%$.

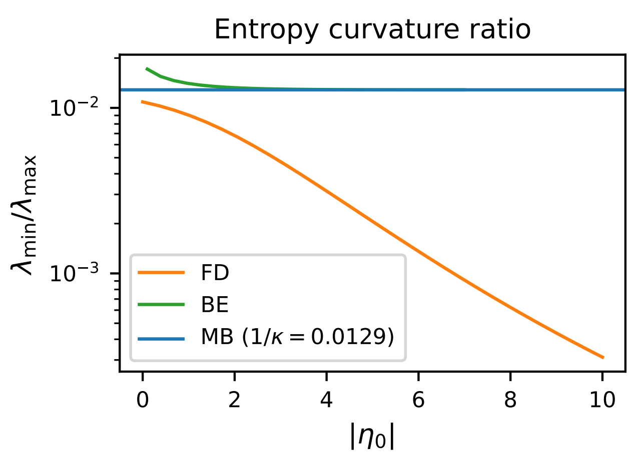

FIG. 3. Entropy curvature ratio $\lambda_{\min}/\lambda_{\max}$ of the Gram matrix versus $|\eta_0|$ at $\Theta = 1.2$. The ratio is the reciprocal of the condition number. MB: constant. FD: decreases (soft direction flattens). BE: above MB throughout. The ratio remains strictly positive for all admissible parameters (Corollary 2).